

Anurag Pradhan

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Portfolio | LinkedIn | GitHub | Google Scholar

SUMMARY

Final-year Computer Science student at VIT Chennai with research experience in Deep Learning, Medical AI, and Generative Models. Currently a Summer Research Intern at IIT Mandi under Prof. Aditya Nigam, working on MRI-to-synthetic CT reconstruction using GANs, VAEs, and diffusion models. Skilled in CNNs, Transformers, LLMs, and VLMs using PyTorch. Co-authored a paper published at IJCNLP–AAACL 2025.

EDUCATION

- Vellore Institute of Technology** Chennai, India
Bachelor of Technology in Computer Science; CGPA: 7.77/10 August 2023 – June 2027
- Sri Venkateswara Jr. College** Visakhapatnam, India
Class 12th (MPC); Score: 88.4/100 July 2021 – June 2023
- Sri Chaitanya Techno School** Visakhapatnam, India
Class 10th Score: 93.6/100 February 2020 – March 2021

SKILLS SUMMARY

- Programming Languages:** Python, C, C++, Java, Kotlin
- Core Expertise:** Vision-Language Models (VLMs), Large/Small Language Models, EdgeAI & On-Device Inference, Agentic AI, Generative AI, Computer Vision, Transformers
- Frameworks & Libraries:** PyTorch, TensorFlow, Hugging Face Transformers, llama.cpp, SmolVLM2, BEiT, BLIP, OpenCV, LangChain
- Tools & Platforms:** Git, Docker, CUDA, ONNX, Weights & Biases, Hugging Face Hub, Android/Compose (JNI), ARM NEON/SVE
- Research Areas:** LLM Weight Quantization, VLM Accessibility Evaluation, Action Recognition, EdgeAI
- Soft Skills:** Leadership, Technical Writing, Event Management, Time Management

EXPERIENCE

- Indian Institute of Technology Mandi (IIT Mandi)** Mandi, Himachal Pradesh
Research Intern May 2026 – June 2026
 - Contribution:** Conducted research in medical imaging under the guidance of Dr. Aditya Nigam, focusing on MRI-to-synthetic CT (sCT) reconstruction for radiotherapy planning and disease detection. Explored and evaluated generative deep learning approaches including VAEs, GANs, and diffusion models for cross-modal image synthesis.
 - Tech Stack:** Deep Learning, Generative Models, VAE, GAN, Diffusion, Vision Encoders
- National Aluminium Company Ltd (NALCO)** Damanjodi, Odisha
Computer Science Intern June 2025 – July 2025
 - Contribution:** Worked on RespiGuard AI, an AI-based respiratory safety system for mining environments. Improved hazard detection reliability using deep-learning-based visual analysis.
 - Tech Stack:** Custom YOLOv5, PyTorch, Data Augmentation, Model Evaluation
- Team Ignition, VIT Chennai** On Site
Flight Software Lead Apr 2024 – Apr 2026
 - Contribution:** Developed flight software for model rockets, including fin actuator control, stability algorithms, onboard video stabilization, and Kalman-based state estimation.
 - Tech Stack:** Arduino, C/C++, PID Control, Image Processing, Deep Learning

PROJECTS

- DrishtiAI — Android App:** Architected an offline multimodal Android assistant using SmolVLM2 and llama.cpp for on-device visual question answering. Built optimized JNI/C++ pipelines for real-time image processing and integrated YOLO-World with Depth-Anything for object-aware depth estimation and spatial reasoning.
- CMCA-ActionNet – Spatio-Temporal Multimodal Action Detection:** Engineered a multimodal human action recognition system achieving 84% accuracy on the JHMDB dataset. Developed a Cross-Modal Causal Attention (CMCA) module to fuse RGB, optical flow, and MediaPipe pose landmarks. Incorporated bidirectional temporal attention and an uncertainty-aware prediction head for robust action recognition and confidence estimation.
- Brain Tumor Segmentation:** Built a BEiT-transformer segmentation pipeline with skip connections and attention-driven decoding. Trained with Albumentations and CE–Dice hybrid loss for stable convergence. [\[GitHub Repo\]](#)
- JanaSathi: Odia E-Governance Chatbot:** Designed a bilingual RAG-based chatbot for Odisha government schemes using OpenRouter LLMs and MiniLM embeddings. Integrated Odia translation with Cohere Command-R+. Achieved 2nd place in the AMD-sponsored OdiaGenAI Hackathon 2025. [\[GitHub Repo\]](#)

CONFERENCE PUBLICATIONS

- Towards Blind and Low-Vision Accessibility of Lightweight VLMs and Custom LLM-Evals:** S. S. Baghel, Y. P. S. Rathore, A. Pradhan. Accepted at the 14th International Joint Conference on Natural Language Processing & the 4th Asian Chapter of ACL (IJCNLP–AAACL 2025), Mumbai, India. [\[Paper Link\]](#)